

José María Barbeito

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EDUCATION

North Carolina State University, *Raleigh, NC*

- **B.S.E:** Chemical Engineering; **Minor:** Material Science Engineering *Expected May 2024*
- **Masters:** Material Science Engineering *Expected May 2025*
- **Current Cumulative GPA:** 3.7/4.0

INDEPENDENT PROJECT EXPERIENCE

Aquarium and Vivarium Construction, Fabrication Engineer, *Durham, NC*

March 2018 - Present

- Constructed a 55 gallon poison dart frog vivarium to house five distinct varieties of poison dart frogs, designed to be a self-sustaining closed cycle.
- Applied wet lab experience to create copper sulfate crystal structures at home.

Arduino Projects

August 2021- Present

- Built a self-controlling evaporator to dry fruit controlled by an Arduino, including a power switch, a heat bulb, and temperature and humidity sensors.
- Constructed various counters and clocks with LED screens and controlled by various mechanical inputs

Liquid Bromine Distillation

December 2021

- Distilled .98 moles of liquid bromine from sodium bromide using a variety of acids, reagents, and reactants. Utilized proper safety procedures and PPE.

WORK EXPERIENCE

Doble W Servicios, *Buenos Aires, Argentina*

February 2021 – March 2021

- Disassembled and reassembled various cars and trucks to install bulletproof glass and kevlar plating for the Argentine Police - averaging ten cars a day fully outfitted.
- Constructed bulletproof glass through assembling and kilning multiple layers of glass and specialty polycarbonate before sanding and installing.

Freelance Contractor, *Durham, NC*

August 2019 – Present

- Worked locally to design and execute a variety of projects, depending on client needs.
- Projects include barn doors, chicken coops, garden fences, bunk beds, benches, tables, electrical installations, bathroom tiling, porch renovations, and concrete repair.

LEADERSHIP AND COMMUNITY EXPERIENCE

TSDCH Chemistry Club, Leader and Co-founder, *Durham, NC*

August 2019 – April 2020

- Founded and led the Chemistry Club at TSDCH.
- Organized a program for first graders to learn about phase changes through super-saturated solutions and gas-producing reactions.

BEE Educated, Founder, *Durham, NC*

August 2014 – Dec 2020

- Founder and lead instructor of BEE Educated, a local outreach program that teaches children ages 8-12 the importance of bees through an on-site program with active apiaries.
- Helped local beekeepers with their own hives, teaching and assisting them in keeping healthy colonies, along with providing swarm capture services for locals.

TECHNICAL SKILLS: Woodworking, Metalworking, Lathe, Crystalline Structure Fabrication, Drone/Digital Cinematography

SOFTWARE SKILLS: JavaScript, Adobe Lightroom, Adobe Photoshop, Video Editing and Production

LANGUAGES: Spanish (Native), German (Beginner)

INTERESTS: Computer Assembly, Backpacking, Traveling, Rock Climbing, Scuba Diving, Drums, Guitar, Crystal Growing